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Assignment: Finding a sufficient Status LED array option that meets engineering and science constraints while meeting requirements from URC

Date: Monday, September 28th

Tldr; Option 1 is much less complex than option 2 but does require precise timing. It is also the cheapest option and would work as long as we implement the sunshade.

- **Option 1:** the WS2815 16×16 matrix, is the cheapest and most flexible design, at about \$124–144 (or around \$95 with a purchased 8" traffic-signal visor instead of a custom hood). It has the widest viewing angle (120°), and any color or pattern can be changed in software. Its worst tradeoff is that blue falls below target, at 56 cd against the 80 cd goal and 2,200 nits against the 4,500-nit goal. It also needs the most control parts: a regulator, a level shifter, SPI setup on the Jetson, and a firmware current cap and requires very precise timing for LED toggling.
- **Option 2:** the LDD-700H LED strings, has the brightest blue (about 260 cd) and passes every target with the simplest wiring: no regulator, level shifter or MOSFETs, and only 0.46 A drawn from the pack but has a more complex mechanical mounted, for about \$214.72. The drawback is a narrow beam that drops to half brightness about 16° off-center, so followers off to the side see much less, and it fails "on" if a control wire comes loose or while the Jetson boots.
- **Option 3:** the Whelen ION lighthead, is the lowest-risk design for daylight visibility, since the heads are built for emergency vehicles, weatherproof and need no regulator. Buying lighthead however is the most expensive option, at about \$384–555, roughly five times the other options, and its brightness can't be confirmed until Whelen provides photometric data and confirms a steady-on mode.
- All three estimates assume the team already has 18 AWG power wire (add \$38.95 if not) and exclude shipping.

Requirements:

1. Rule [1.f.vi](#): "There must be an LED indicator on the back of the rover, visible in bright daylight (e.g. LED array or high power LED), that will signal:
 - a. Red: Autonomous operation
 - b. Blue: teleoperation (manually driving)
 - c. Flashing green: Successful arrival at target"
2. Rule 2.b.i: "All of the competition events will be held in full daylight"
3. Rule 3.a.v: The E-stop "shall immediately stop the rover's movement and cease all power draw from any and all batteries...All systems on the rover must have the power switched off by this single E-stop"
4. The PDB's 12 V outputs are not regulated. A 4S6P LiFePO₄ pack runs about 10-14.6 V (12.8 V nominal, 36 Ah). The only regulator on the diagram is the Jetson's S13V25F12
5. The controller is the Jetson Orin Nano. Its 40-pin header uses 3.3V logic and also has 5V pins.
6. A mount that protects from dust, rain, and poor weather conditions

Targets:

1. Luminance: how bright the lit surface is, targeting > 1500 nits in direct sunlight = 4500 cd/m² for outdoors
2. Intensity: how visible the light is from a distance, where it looks like a dot, targeting > 80 cd per color on-axis based on the US minimum for a car brake light (FMVSS 108) which is designed to be seen in daylight and similarly taillights are 2-18 cd, which are designed to be seen well in nighttime
3. Visibility to the eye: candelas (cd) accounts for this as it is a measure based on sensitivity of the human eye to each color, but to note blue is the hardest color to make bright

Option 1: WS3815 16x16 addressable matrix

Design

The panel gets its power from a PDB output. That line goes through a 5 A inline fuse to a Pololu S13V25F12 regulator, whose steady 12 V output feeds the



panel's +12V and ground wires. A 1000 uF, 25 V capacitor sits across the panel's power input. For control, the Jetson sends color data from its SPI MOSI pin. The signal passes through a 74AHCT125 level shifter, then a 330 ohm series resistor, into the panel's DIN. The panel's backup input (BIN) is tied to ground. Essentially the level shifter is required because it is sending timed data to the LEDs, not just toggling LEDs like in normal embedded systems. The WS3815 reads data from one wire called CIN in pulses (800000 bits per second -> 1.25 us per bit). The panel measures each pulse to be able to tell from high or low (high pulse is 0.3 us, active low, and long pulse is 0.8 us which means 1). Each pixel then has 24 bits (8 for green, 8 for red, 8 for blue) and then there are 256 total pixels. Since the Jetson uses SPI, Linux can't reliably toggle a pin with a sub-microsecond precision, so the SPI hardware generates pulses instead. Each LED bit becomes 3 SPI bits (100 makes a short pulse, a 0, and 110 makes a long pulse, a 1). Each pulse then has to be read correctly, thousands of times a second. A 3.3 V signal is already below the panel's 3.5 V threshold, and each edge takes time to rise. So a pulse can look shorter than it is, or not register at all, and a single misread bit shifts the colors on every pixel after it. The shifter's full 5 V output makes each pulse clearly readable.

[Click here for an AI visual in Figma:](#) [📄 Status LED – Option 1 Layout & Wiring](#)

[Click here for draw.io for connections:](#) [StatusLEDOption1.drawio](#)

Weather Protection

A sealed, dust-tight and waterproof Hammon box with a clear lid(1554W2GYCL,180 x 180 x 60mm. I would be concerned about the heat. With a sunshade, the heat should be reduced, but still inside of the box at 2A current which is about 24 W, that is about 30 degrees C inside the box before the sun. Figure on the box warming by 1.3 degrees C per watt. My main concern with the heat is the operating ranges for each of the electronic parts. So for the LED panel, the rated operating range os -25 degrees C to 85 degrees C, the level shifter has -40 deg C to 85 deg C. Electrolytic capacitor rated for 85 deg C to 105 deg C. The most-at risk part is the panel and level shifter since its limit is 85 deg C. Inside the box, it sees 35 deg from the desert air, 30 deg C at most from the LEDs, plus light it sees from the sun. That's a minimum of 65 deg C which is fairly cose to the 85 deg C so this def shows we will need a sun shade going with this option.

Parts List

Part	Qty	Datasheet	Purchase Link	Unit cost	Total
WS2815 16x16 matrix, 12 V	1	https://www.norandled.com/upload/201808/WS2815%20LED%20Datasheet.pdf	WS2815 RGB LED Matrix Panel DC12V Addressable Dual Signal Display – BTF-LIGHTING	\$26.10	
1554W2GYCL Hammond enclosure (180 x 180 x 60 mm)	1	NEMA 4, 4X, 6, 6P, 12, and 13 Polycarbonate Miniature Cases with Flat Lid AutomationDirect	Enclosure: 180 x 180 x 60mm, polycarbonate (PN# 1554W2GYCL) AutomationDirect	\$33	
Pololu S13V25F12 regulator	1	12V, 2.5A Step-Up/Step-Down Voltage	12V, 2.5A Step-Up/Step-Down Voltage	\$18.95	

		Regulator S13V25F12	Regulator S13V25F12		
74AHCT125 level shifter / level shifter (14-pin DIP chip)	1	SNx4AHCT12 5 Quadruple Bus Buffer Gates With 3-State Outputs datasheet (Rev. R)	Quad Level-Shifter - 74AHCT125 - (3V to 5V) (D96)	\$2.95	
Inline ATO/ATC fuse holder, 18 AWG, 5A fuse included	1		Inline ATO/ATC Fuse Holder – 18AWG	CAD \$2.15	
1000 uF 25V electrolytic capacitor	1		Electrolytic Decoupling Capacitors - 1000uF/25V - SparkFun Electronics	\$0.50	
PG-9 cable gland (seals where the 12 V power cable enters the box)	1		https://www.ad fruit.com/pro duct/761	\$1.95	
PG-7 cable gland (seals where the data cable enters the box)	1		https://www.ad fruit.com/pro duct/762	\$1.95	
Aluminum backing plate, about 164 x 164 x 3 mm (holds the flexible panel flat and spreads its heat)	1		Team stock/generic	~\$10	
aluminum standoffs, about 42 mm (M3 or M4) to hold the panel 3-5 mm under the lid and carry the heat	4		generic	~\$5	

to the box floor					
Mounting hardware” stainless screws, sealing washers, rubber isolators, thread-locker (bolts the box to the rover and seals the screw holes)	1 set			\$5-10	
Jacketed round cable: 2-conductor 18 AWG (power) and 2-conductor 22-24 AWG (data + GND) (glands only seal around a round jacketed cable, loose single wires will leak)	1 m each			\$5-10	
M12 screw vent (to stop the lid fogging from condensation)	1 2-pack		Dusproof Waterproof IP 68 Lighting Breather Screw Air Vent Pressure Reduce Valve M12x1.5 Venting Screw Breather Plug - 2Pack	\$8.49	
330 ohm ¼ W resistor, 20-pack (data-line series resistor)	1 pack	Resistors - SparkFun Learn	Resistor 330 Ohm 1/4 Watt PTH - 20 pack (Thick Leads) - SparkFun Electronics	\$1.25	
22 AWG stranded wire, 25ft (data, 5V and ground to the shifter)	1	80 deg C/ 300 V rating (on product page)	Stranded-Core Wire Spool - 25ft - 22AWG - Black: Adafruit	\$2.95	

			Industries. Unique & fun DIY electronics and kits		
Sunshade	1	Description above for CAD		\$0-\$10	
18 AWG power wire (PDB -> fuse -> regulator -> panel)	-	https://www.jameco.com/z/1MS822-01G-18-AWG-6-Color-Stranded-Tinned-Copper-Hook-Up-Wire-Assortment-25-Feet_2206281.html	Team already owns. The fuse holder includes 300 mm of 18 AWG. If needed, Jameco 6-color 18 AWG assortment	\$0 - \$38.95	
Jetson Orin Nano (controller)	-	-	-	-	-
Items before custom CAD + mounting					\$77.85
Adding CAD + mounting					\$124.75-\$144.75

Targets

Color	Intensity/LED, Total (cd)	Luminance	> 80 cd	> 1500 nits	> 4500 nits
Red	0.36, 92	3600	✓	✓	✗
Green	1.15, 294	11500	✓	✓	✓
Blue	0.22, 56	2200	✗	✓	✗
White (ref)	1.71, 438	17100	✓	✓	✓

Notes on targets:

- Visibility: the 120 degree viewing angle passes and the 16 x 16 cm area is larger than 8 cm x 8 cm which was originally a requirement in 2022 (no longer a requirement but a reference for sizing)
- Overall, the red and green are sufficient and blue is marginally covered, so this design would work best with a sunshade
- The 8 cm x 8 cm fails for all three

Sunshade

- We should have a mechanical team make a sunshade. The ProAm7” hood’s opening is 115 mm tall, which is too short for the 160 mm panel. We should have a cap-styl hood with 170 x 170 mm interior, 80-100 mm deep and open at the bottom. An 8” traffic-signal visor also works but shades the panel’s corners.
- A clear polycarbonate cover would suffice for protecting against the dust. The panel isn’t sealed.

Features

Feature	Part
Red	Yes
Blue	Yes
Green	Yes
Flashing green	Yes
Size	16 cm x 16 cm
Input voltage	12 V DC (datasheet range 9.5-13.5 V)
LEDs	256 RGB LEDs (5050)
Individually addressable	Yes
Viewing angle	120 deg
Power	0.3 W/LED max
Weatherproof	No, IP20-IP30
Price	\$26.10
Wavelength	Red 620-625 nm, Green 515-525 nm, Blue 465-475 nm
Package includes	1 panel and a 4-pin connector. No power supply or controller

Power Calculations

- Max power: $256 * 0.3 \text{ W} = 76.8 \text{ W}$
 - $I = 76.8/12 = 6.4 \text{ A}$ at full white LED
- The S13V25F12 is rated for 2.5 A. The current must be limited to 2.0A in firmware.
- Only one color should ever be lit. If one color draws one third of full white (about 2.1 A), it runs at nearly full brightness under that cutoff. This should be measured on the bench in application.
- Current drawn from the battery at 24 W output: $24 \text{ W} / (0.9 * 12.8 \text{ V}) = 2.1 \text{ A}$. At a low 10 V pack, it rises to 2.7A

Concern: Is a voltage regulator required?

- It is required. The pack is 8.8-14.6 V, which goes both above and below the panel’s 9.5-13.5V range.
- Recommending Pololu S13V25F12 voltage regulator
- It accepts 2.8-22V in and outputs 12 V +/- 4% (11.5-12.5V), which is inside the panel’s range
- It is a buck-boost regulator, so it holds 12 V whether the pack voltage is above or below 12 V
- It is rated for about 2.5 A, but the real limit depends on heat so it must receive airflow

- This is the same regulator the Jetson uses

Pin	Voltage
Regulator Vin	8.8-14.6 V (pack)
Regulator, Vout, panel +12V	11.5-12.5 V
Jetson SPI MOSI	3.3V (logic)
24AHCT125 VCC	5V from Jetson 5V header pin, which the chip allows 4.5-5.5 V
74AHCT125 input	3.3 V is read as high (threshold is 2.0 V)
74AHCT125 output, panel DIN	about 5 V; the panel needs at least 3.5 V, so the shifter is required
74AHCT125 /OE and unused inputs	GND
Panel BIN	GND

Voltage at each pin. The shifter is required because the Jetson's data signal is too low for the panel to read reliably. The panel needs at least 3.5 V to read 1, as the WS 2815 chip in each pixel runs its logic at 5V internally and its datasheet sets the logic-high threshold at $0.7 \cdot V_{DD} \rightarrow 0.7 \cdot 5 = 3.5V$. **With this in mind, the Jetson only outputs 3.3 V.** Its 40-pin header uses 3.3 V logic, and that 3.3 V is the best case, which is 0.2V short of what the panel requires. **They might work without the shifter but why take that chance?**

Wiring

Run	Gauge
PDB to fuse to regulator to panel (+ and -)	18 AWG
Data (shifter to DIN) and 5V/GND to the shifter	22-24 AWG; keep data under 1 m, reliable to 10 m with a shifter

Control

- Enable SPI on the 40-pin header with jetson-io. Pin 19 is MOSI
- SPI should be run at 2.4 MHz and send 3 SPI bits per LED bit. One frame is 2304 bytes.
- Hold the data line low between frames for the reset time the datasheet specifies.
- Only resend a frame when the rover's state changes. Flashing green is alternating a green frame and off frame at 1-2 Hz.

Option 2: Mean Well LDD-700H single-color high-power LED strings

Design

The LEDs get their power from a PDB output. That line goes through a 5 A inline fuse to the inputs of three Mean Well LDD-700H constant-current drivers wired in parallel. Each driver powers one color string of two Cree XP-E2 LEDs in series at 700 mA: one red, one green and one blue string. All six LEDs are mounted on an 80 x 80 mm aluminum plate that spreads their heat, each under a 20 mm Carclo wide-spot lens. For control, each driver's DIM pin connects directly to one Jetson GPIO pin, with a 10 kohm pulldown resistor from DIM to ground. No voltage regulator, level shifter, MOSFETs or LED resistors are needed.

XLamp® XP-E2 LEDs



AI Visual with Figma: [📄 Status LED – Option 1 Layout & Wiring](#)

Draw.io here for power and controls connections: [statusled option2.drawio](#)

Parts List

Part	Qty	Datasheet	Purchase Link	Unit cost	Total
Cree XP-E2 LED star, red (620-630 nm), 20 mm board	2	Cree XLamp XP-E2 LEDs datasheet	Cree XLamp XP-E2 Color High Power LED Star - LEDSupply	\$5.25	\$10.50
Cree XP-E2 LED star, green (520535 nm), 20 mm board	2	Cree XLamp XP-E2 LEDs datasheet	Cree XLamp XP-E2 Color High Power LED Star - LEDSupply	\$5.25	\$10.50
Cree XP-E2 LED star, blue (465485 nm), 20 mm board	2	Cree XLamp XP-E2 LEDs datasheet	Cree XLamp XP-E2 Color High Power LED Star - LEDSupply	\$5.25	\$10.50
Mean Well LDD-700H constantcurrent LED driver	3	LDD-H series datasheet - MEAN WELL	MEAN WELL LDD700H DC-DC Converter, 52V, 0.7A, 9-56Vin, LED	\$4.30	\$12.90

			Driver - RS		
Carclo 10196 20 mm frosted widespot lens	6	20.0mm Wide Spot Frosted TIR (10196) documentation	10196 Carclo Lens - Frosted Wide Spot LED Optic - LEDSupply	\$1.81	\$10.86
5052-H32 aluminum sheet, 0.063" × 12" × 18" (LED plate + rover bracket)	1		https://www.spedymetals.com/ps-1939-133-0063-5052-h32-aluminum-sheet.aspx	\$13.18	
Luxeon Star LXT-S-12 thermal tape, 20 mm hex (12 pads)	1		https://www.luxeonstar.com/lxt-s-12	\$7.49	
Hammond 1554NPL inner panel (galvanized steel, sizes N and P)	1	1554NPL - Hammond Mfg.	Hammond 1554NPL Panel for 1554N, P - Galv TEquipment	\$10.74	
Hammond 1554FT mounting feet (4 feet + stainless screws and nuts)	1	1554FT - Hammond Mfg.	1554FT HAMMOND Kit, 4 Feet, 4 Stainless Steel Screws & Nuts Newark Electronics	\$9.83	
Pololu 4-40 aluminum standoff, 1.25", M-F, 4-pack (#1951)	1		https://www.pololu.com/product/1951	\$4.95	
Pololu 4-40 aluminum standoff, 1", M-F, 4-pack (#1950)	1		https://www.pololu.com/product/1950	\$4.70	

Pololu 4-40 aluminum standoff, 3/4", M-F, 4-pack (#1949)	1		https://www.pololu.com/product/1949	\$3.85	
Pololu 4-40 aluminum standoff, 1/2", M-F, 4-pack (#1948, driver board)	1		https://www.pololu.com/product/1948	\$3.33	
Adafruit Perma-Proto half-size, 3-pack (#571, drivers + pull-downs)	1		https://www.adafruit.com/product/571	\$12.50	
#4-40 × 1/4" stainless pan Phillips screw	10		https://boltdepot.com/Product-Details?product=5312	\$0.06	
#4-40 stainless nylon-insert lock nut	10		https://www.boltdepot.com/Product-Details.aspx?product=4107	\$0.06	
M3 × 8 mm stainless pan Phillips screw (inner panel to box inserts)	4		https://boltdepot.com/Product-Details?product=6831	\$0.05	
#10-24 × 3/4" stainless pan Phillips screw (feet to bracket, bracket to frame)	6		https://www.fastener-warehouse.com/Phillips-Pan-Machine-Screw-188-Stainless-Steel-p/10001224.htm	\$0.08	

Resistor kit, ¼ W, 500 pcs (10 kohm DIM pulldowns)	1	Resistors - SparkFun Learn	Resistor Kit - 1/4W (500 total) - SparkFun Electronics	\$9.95	\$9.95
Inline ATO/ATC fuse holder, 18 AWG, 5A fuse included	1		Inline ATO/ATC Fuse Holder – 18AWG	~\$2.15 (CAD \$2.95)	~\$2.15

22 AWG stranded wire, 25 ft (driver to LEDs, DIM signals, ground to Jetson)	1	80 deg C/ 300 V rating (on product page)	StrandedCore Wire Spool - 25ft - 22AWG - Black : Adafruit Industries. Unique & fun DIY electronics and kits	\$2.95	\$2.95
ProAm USA 7" LCD monitor hood (sunshade)	1	Dimensions on product page	7 inch LCD Video Monitor Hood / Sunshade by ProAm USA	\$19.99	\$19.99
Aluminum plate, 80 x 80 x 3 mm, + thermal adhesive (LED heat spreader)	1	None	Team stock or generic	~\$10	~\$10
Clear UVcoated polycarbonate sheet, 8x10x0.093 in (dust cover)	1	None	LEXAN 8 in. x 10 in. x 0.093 in. Clear UV Coated Polycarbonate Sheet 31GE-XL-1 - The Home Depot	\$10	\$10
18 AWG power wire (PDB -> fuse)	-	https:// www.jameco.com/z/	Team already owns. The fuse holder	\$0 - \$38.95	

-> drivers)		JMS822-01G-18AWG-6-ColorStranded-TinnedCopperHook-Up-WireAssortment-25-Feet_2206281.html	includes 300 mm of 18 AWG. If needed, Jameco 6color 18 AWG assortment		
Jetson Orin Nano (controller)	-	-	-	-	-
Total					\$214.21

Targets

Color	Intensity/ LED, Total (cd)	Luminance (nits)	> 80 cd	> 1500 nits	> 4500 nits
Red	285, 570	907,000	✓	✓	✓
Green	455, 910	1,449,000	✓	✓	✓
Blue	130, 260	414,000	✓	✓	✓
White (ref)	N/A (no white channel)	-	-	-	-

- Notes:
 - How the values were found: flux per LED at 350 mA (LEDSupply listing) is red 67.2 lm, green 107 lm and blue 30.6 lm. At 700 mA, Cree's relative-flux curve gives about 1.7x that: red 114 lm, green 182 lm and blue 52 lm.
 - The Carclo 10196 lens datasheet gives a peak intensity of 2.50 cd/lm (78% efficient, measured with the Cree XP-E). Intensity per LED = flux x 2.50, and the total is 2 LEDs per color
 - Luminance = intensity / lens area. Two 20 mm lenses = $2 \times \pi \times (0.01 \text{ m})^2 = 6.28 \times 10^{-4} \text{ m}^2$
 - Visibility: the lens beam is 31.5 deg wide (FWHM), so the light drops to half brightness at about ± 16 deg off-axis. Even at half brightness, blue is still 130 cd. **At 10 m behind the rover, ± 16 deg covers about ± 2.8 m to either side.** This is a much narrower cone than Option 1's 120 deg.

- Each color is two 20 mm lenses, smaller than the 8 cm x 8 cm reference from 2022. Spread the six lenses across the 80 mm plate so the lit pattern covers that area.
- Do not add a diffuser to make it look bigger as it spreads the light back out and blue drops to 33 cd which would fail
- Without lenses, bare LEDs (130 deg beam) give red 73 cd, green 116 cd and blue 33 cd total, so red and blue fail; lenses are required
- Overall, all three colors pass each target and blue is the brightest of the three options. The tradeoff is the narrower viewing cone. The lens values were measured with the Cree XP-E (same 3.45 mm package as the XP-E2)
- Sunshade
 - The ProAm USA 7" monitor hood (\$19.99) fits this design. Exterior 18 x 12.7 cm, interior 17 x 11.5 cm, depth 12.7 cm at the top tapering to 9.5 cm at the bottom. It is nylon over a rigid frame, attached with adhesive Velcro and the bottom piece is removable. The 80 x 80 mm plate fits inside with room.
 - A clear polycarbonate cover would suffice for protecting against the dust. The LED plate isn't sealed, but the drivers are fully encapsulated (IP67)

Features

Feature	Part
Red	Yes (Cree XP-E2 red x2)
Blue	Yes (Cree XP-E2 blue x2)
Green	Yes (Cree XP-E2 green x2)
Flashing green	Yes, software controlled (Jetson toggles the DIM pin)
Size	20 mm LED boards on an 80 x 80 mm plate; each driver is 31.8 x 20.3 x 12.2 mm
Input voltage	Driver: 9-52 V DC, runs straight from the pack
LEDs	6 high-power LEDs (2 per color)
Individually addressable	No (3 on/off channels)
Viewing angle	130-135 deg bare; 31.5 deg (FWHM) with the Carclo 10196 lens
Power	Up to 3.6 W red, 4.9 W blue, 5.3 W green (one color lit at a time)
Weatherproof	Drivers yes (IP67 encapsulated); LED plate no, needs the cover
Price	\$5.25 per LED; \$4.30 per driver; \$1.81 per lens

Wavelength	Red 620-630 nm, Green 520-535 nm, Blue 465-485 nm
Package includes	Each LED comes on a 20 mm aluminum star board. Driver sold alone. Lenses, holders, plate and wire sold separately

Power Calculations

- **Worst-case color is green:** $P = 2 \text{ LEDs} \times 3.8 \text{ V} \times 0.7 \text{ A} = 5.3 \text{ W}$ at the LEDs
 - Assuming about 90% driver efficiency at a 12.8 V input: $5.3 / 0.9 = 5.9 \text{ W}$ from the pack
 - Red: $2 \times 2.6 \text{ V} \times 0.7 \text{ A} = 3.6 \text{ W}$
 - Blue: $2 \times 3.5 \text{ V} \times 0.7 \text{ A} = 4.9 \text{ W}$
 - Current drawn from the battery: $I = 5.9 / 12.8 = 0.46 \text{ A}$. At a low 9 V pack, it rises to 0.66A
- Only one color should ever be lit, so the aluminum plate only has to shed 3.6-5.3 W
- The 5 A fuse included with the holder protects the 18 AWG wire. A 3 A fuse is closer to the load if you want tighter protection
- Energy: 5.9 W for one hour = 5.9 Wh, about 1.3% of the 460 Wh pack

Voltage Regulation

- It is not required. The LDD-700H is itself a regulator: it holds the LED current at 700 mA anywhere across its 9-52 V input range
- The pack is 8.8-14.6 V, which is inside that range except for the last 0.2 V before the BMS cuts power at 8.8 V
- The LED strings use at most 5.2 V (red) and 7.6 V (green), well below the pack's typical 12.8 V, which leaves the driver plenty of headroom
- Near the bottom of the pack (about 9 V), the green string only has about 1.4 V of headroom and may dim slightly. That only happens right before the BMS shuts off anyway
- No regulator also means one less part to overheat or fail

Voltages

Pin	Voltage
Driver +Vin / -Vin	8.8-14.6 V (pack) / PDB ground
Driver +Vo / -Vo, red string	About 4.4-5.2 V at 700 mA
Driver +Vo / -Vo, green string	About 6.4-7.6 V at 700 mA
Driver +Vo / -Vo, blue string	About 6.2-7.0 V at 700 mA
DIM from Jetson GPIO	3.3 V = on (needs 2.5-6 V); 0 V = off (below 0.8 V)
DIM pulldown	10 kohm from DIM to -Vin (GND)
Jetson GND	Tied to driver -Vin (PDB ground / BMS P-)

- The DIM pin turns on at 2.5 V, so the Jetson's 3.3 V output has 0.8 V of margin. **Watch the DIM pin: it is on when left open.** While the Jetson boots, its pins float, so all three colors could light up. The 10 kohm pulldown should hold them off, but test it on the bench. If a DIM wire comes loose, that color turns on (it fails on).

Wiring

Run	Gauge
PDB to fuse to drivers (+ and -)	18 AWG
Driver to LED strings (0.7 A)	20-22 AWG
DIM signals and GND to the Jetson	22-24 AWG

Control

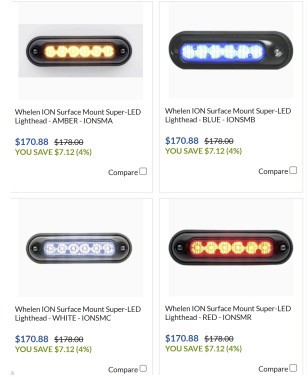
- Use three free GPIO pins on the Jetson 40-pin header, set as outputs (libgpiod or Jetson.GPIO)
- High = that color on, low = off. Red for autonomous, blue for teleoperation
- Flashing green is toggling the green pin at 1-2 Hz
- Drive all three pins low at startup, before the mode logic runsConnect the Jetson ground to the driver ground (PDB ground)

Option 3: Whelen ION vehicle lightheads (red, blue, green)

Design

The lightheads get their power from a PDB output. That line goes through a 5 A inline fuse to the +12 V wire of all three lightheads. Each head's ground wire runs through its own IRLB8721 logic-level MOSFET to the PDB ground. For control, each MOSFET gate connects to a Jetson GPIO pin through a 100 ohm resistor, with a 100 kohm resistor from gate to ground so the light stays off while the Jetson boots. No voltage regulator or level shifter is needed.

AI Visual in Figma:  Status LED – Option 1 Layout & Wiring draw.io here:



Parts List

Part	Qty	Datasheet	Purchase Link	Unit cost	Total
Whelen IONSMR lighthead, red	1	Whelen ION Surface Mount Series Super-LED Lightheads - SWPS	Whelen ION Surface Mount Super-LED Lighthead - RED - IONSMR from SWPS.com	\$170.88	\$170.88
Whelen IONSMB lighthead, blue	1	Whelen ION Surface Mount Series Super-LED Lightheads -	Whelen ION Surface Mount Super-LED Lighthead -	\$170.88	\$170.88
		SWPS	BLUE - IONSMB from SWPS.com		
Whelen IONSMG lighthead, green	1	Whelen ION Surface Mount Series Super-LED Lightheads - SWPS	Whelen ION Series, Surface Mount, Green with Clear Lens - IONSMG	138.39	~\$138.39

IRLB8721 logic-level N- MOSFET (TO-220)	3	IRLB8721PB F datasheet	N-channel power MOSFET - 30V / 60A : Adafruit Industries	\$2.25	\$6.75
Resistor kit, ¼ W, 500 pcs (100 ohm gate, 100 kohm pulldown)	1	Resistors - SparkFun Learn	Resistor Kit - 1/4W (500 total) - SparkFun Electronics	\$9.95	\$9.95
Inline ATO/ATC fuse holder, 18 AWG, 5A fuse included	1		Inline ATO/ATC Fuse Holder - 18AWG	~\$2.15 (CAD \$2.95)	~\$2.15
22 AWG stranded wire, 25 ft (gate signals, ground to Jetson)	1	80 deg C/ 300 V rating (on product page)	StrandedCore Wire Spool - 25ft - 22AWG - Black : Adafruit Industries. Unique & fun DIY electronics and kits	\$2.95	\$2.95
ProAm USA 7” LCD monitor hood (sunshade)	1	Dimensions on product page	7 inch LCD Video Monitor Hood / Sunshade by ProAm USA	\$19.99	\$19.99
18 AWG power wire	-	https:// www.jameco.	Team already owns. The	\$0 - \$38.95	

(PDB -> fuse -> heads, heads -> MOSFETs)		com/z/ JMS822-01G-18AWG-6-ColorStranded-TinnedCopperHook-Up-WireAssortment-25-Feet_2206281.html	fuse holder includes 300 mm of 18 AWG. If needed, Jameco 6color 18 AWG assortment		
Jetson Orin Nano (controller)	-	-	-	-	-
Items with listed prices (red, blue, parts)					\$383.55
Adding green (estimated)					~\$555

Targets

Color	Intensity/ LED, Total (cd)	Luminance (nits)	> 80 cd	> 1500 nits	> 4500 nits
Red	Not published	Not published	✓*	✓*	✓*
Green	Not published	Not published	✓*	✓*	✓*
Blue	Not published	Not published	✓*	✓*	✓*
White (ref)	N/A (no white head)	-	-	-	-

Notes on targets:

- Whelen doesn't publish intensity or luminance on the product page. The heads meet SAE J595, the standard for emergency-vehicle warning lights, which are designed to be seen in daylight.
- Request the photometric data from Whelen before buying.

- Visibility: viewing angle is not published. Each head's lit face is about 3.3 x 10.2 cm (34 cm²), smaller than the 8 cm x 8 cm (64 cm²) reference from 2022. Mounting the three heads side by side makes a 10 x 10 cm block.
- Overall, this is the lowest-risk option for daylight visibility, but it can't be confirmed with numbers until Whelen provides photometrics.

Sunshade

- The ProAm USA 7" monitor hood (\$19.99) fits all three heads side by side (3 x 33 mm = 99 mm wide x 102 mm tall) inside its 17 x 11.5 cm interior. Depth 12.7 cm at the top tapering to 9.5 cm at the bottom; attaches with adhesive Velcro.
- No dust cover is needed. The heads are listed as water-resistant/waterproof and are built for exterior vehicle mounting.

Features

Feature	Part
Red	Yes (IONSMR)
Blue	Yes (IONSMB)
Green	Yes (IONSMG)
Flashing green	Yes, built-in flash patterns (25) or the Jetson toggles it
Size	25 x 33 x 102 mm (1" H x 1-5/16" W x 4" L) each
Input voltage	12.8 V DC nominal (vehicle system)
LEDs	Whelen Super-LED (count not published)
Individually addressable	No (3 separate on/off heads)
Viewing angle	Not published
Power	0.40 A average at 12.8 V (about 5.1 W) per head
Weatherproof	Listed as water-resistant/waterproof; IP rating not stated
Price	\$170.88 each (red and blue listed; green not listed)
Wavelength	Not published
Package includes	Lighthouse with a 6" 4-wire pigtail. Five-year warranty

Power Calculations

- One head: $P = 12.8 \text{ V} \times 0.40 \text{ A} = 5.1 \text{ W}$
 - Only one head should ever be lit, so 0.40 A from the pack
 - If all three were on by mistake: $3 \times 0.40 \text{ A} = 1.2 \text{ A}$, still under the 5 A fuse
- MOSFET loss: $P = I^2 \times R = (0.40 \text{ A})^2 \times 0.016\text{-}0.05 \text{ ohm} = \text{under } 0.01 \text{ W}$, so no heatsink is needed
- Energy: 5.1 W for one hour = 5.1 Wh, about 1.1% of the 460 Wh pack
- The 0.5 A is listed as an average across flash patterns. A steady-on pattern may draw more, so we should measure this.

Is a voltage regulator required?

- It is not required. The heads are built for 12 V vehicle electrical systems, which swing about as much as the 8.8-14.6 V pack does
- Confirm the minimum and maximum input voltage on Whelen's spec sheet. If the minimum is above 10 V, the heads may cut out near the bottom of the pack
- No regulator means no extra heat source and one less part to fail

Pin	Voltage
Lighthouse + (power wire)	8.8-14.6 V (pack)
Lighthouse - (ground wire)	To MOSFET drain; switched to PDB ground
MOSFET source	PDB ground
MOSFET gate	3.3 V from Jetson GPIO through 100 ohm; IRLB8721 turns on at 2.35 V max
Gate pulldown	100 kohm from gate to GND
Pattern / sync wires	Pattern set once to steady-on; sync wire left unconnected and insulated
Jetson GND	Tied to PDB ground / BMS P-

The MOSFET switches the ground wire, so the pattern and sync wires must not be tied to ground anywhere else, or they could bypass the switch and leave the head on.

Wiring

Run	Gauge
PDB to fuse to heads (+)	18 AWG
Head (-) to MOSFET to PDB ground	18 AWG
Gate signals and GND to the Jetson	22-24 AWG

Control

- Use three free GPIO pins on the Jetson 40-pin header, set as outputs (libgpod or Jetson.GPIO)
- High = that head on, low = off. Red for autonomous, blue for teleoperation

- Set each head's pattern once to steady-on. The product page only mentions flash patterns, so we shall confirm the model has a steady-pattern before buying
- Flashing green: toggle the green pin at 1-2 Hz, or use the head's own flash pattern.
- Toggling power may restart the pattern, so test which looks cleaner
- Drive all three pins low at startup before the mode logic runs

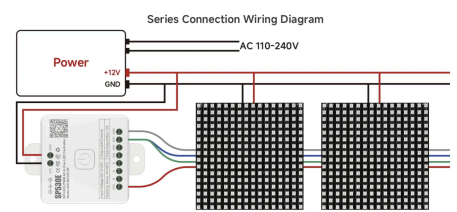
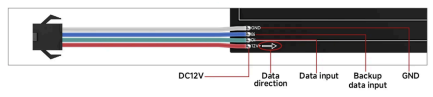
What is option 1: an LED array controlled with a microcontroller using SPI protocol and a logic level shifter. Total cost is \$ 26.10 + \$4.99 + \$2.95 + \$5 for cushion = \$39.04.

Part Options:

Option 1: WS2815 RGB LED Matrix Panel + microcontroller

- Name: WS2815 23 V WS2315 8x8 RGB matrix
- Datasheet: [WS2815 RGB LED Matrix Panel | DC12V Addressable Dual Signal Display – BTF-LIGHTING](#)
- Requires: 12 V Input and SPI Controller

Feature	Part
Red	Yes
Blue	Yes
Green	Yes
Flashing green	Software controlled
Size	8 cm x 8 cm
Input voltage	12V (DC)
LEDs	64 RGB LEDs
Individually Addressable	Yes
Viewing Angle	120 deg
Power	0.3 W/LED theoretical max
Weatherproof	No – IP30
Price	\$10.26 or \$26.10 if sizing up
Wavelength (nm)	Red 650 nm, Green 520 nm, Blue 460 nm
Package includes	• 1 x WS2815 DC 12 V Panel Screen • 1 x 4 Pin connector • Does not include LED Power and SPI Led Controller

Series Connection Wiring Diagram	
	
How much current does it need?	0.3 W / LED With 64 LEDs: $P_{\max} = 64 * 0.3 = 19.2 \text{ W}$ At 12 V: $I = P/V$ $I = 19.2 / 12$ $I_{\max} = 1.6 \text{ A}$

*Part 1 would also need a Controller. The manufacturer recommends an arduino, teensey, or something of that nature

Brightness calculations: (on its own sub tab of this)

1. [How Many Nits Do You Need to See a Screen in Direct Sunlight? Outdoor Brightness Guide\(2026\)](#)
 - a. You need at least 1500 nits to see a screen clearly in the sun. For shaded outdoor areas, 1000 nits or more is usually enough
2. Outdoor and Semi-Outdoor applications require a minimum brightness of 4500 cd/m²
 - a. Covered areas or storefront windows need more of a brightness range of 2000-4000 cd/m²

Microcontroller:

- Option 1 – Raspberry Pi Pico 2 with headers plus a 74AHCT125 Logic Level Shifter
 - Name: Raspberry Pi Pico 2 with headers
 - Cost = \$4.99
 - Features:

- Dual ARM Cortex M33 Processor
 - 520KB RAM
 - 150 MHz clock rate
 - SDRAM 520 KB
 - With a higher core clock speed, double the memory, more powerful Arm cores, optional RISC-V cores, new security features, and upgraded interfacing capabilities, Raspberry Pi Pico 2 delivers a significant performance boost, while retaining compatibility with earlier members of the Raspberry Pi Pico series.
- Link: [Raspberry Pi Pico 2 with Header; Dual ARM Cortex M33 Processor; 520KB RAM - Micro Center](#)
- Name: 74AHCT125 Logic Level Shifter
 - Cost = \$2.95
 - Link: [https://vetco.net/products/vupn524_quad_level-shifter - 74ahct125 - 3v to 5v d96?variant=42879377801353&gsid=ePoodpKosXHc&utm](https://vetco.net/products/vupn524_quad_level-shifter_-_74ahct125_-_3v_to_5v_d96?variant=42879377801353&gsid=ePoodpKosXHc&utm)
 - This quad 74AHCT125-based level shifter is a practical choice for converting 3.3 V logic signals to reliable 5 V logic, making it well suited for microcontroller and LED data applications. Its use of the widely supported 74AHCT125 family makes it attractive for hobbyists and embedded projects that need clean signal translation rather than complex bidirectional level shifting.
 - Community discussions around 74AHCT125-based level shifters are broadly positive when the chip is used for the jobs it was designed for, especially driving 5 V logic inputs such as addressable LEDs. Reviewers consistently emphasize that success depends on using it for logic signals—not power conversion—and selecting it for compatible one-way level-shifting applications.
 - Technical discussions note that while the 74AHCT125 is effective as a buffer and level shifter, designers should ensure it is used appropriately for signal buffering and voltage translation rather than as a universal interface.
 - Features:
 - Output current +/- 8 mA at 5V
 - Max propagation delay: 5ns at 5V
 - Low power consumption: typical ICC of only 1 uA
 - Operating voltage 2V to 5.5V

Tldr;

I would design this as a 12 V RGB addressable matrix with a small dedicated controller rather than power a 5 V NeoPixel matrix directly from the Jetson

Title: Status LED

Assigned to: Jennifer Durnall

-
- Research the power regulator if needed (step-up or step-down)
 - Look at the brand for the jetson
 - See if the Status LED array can run on 12 V (preferable) and look at the current ratings as well for the power regulator)
 -

Source: [URC2022 Q&A](#) ** FROM 2022 NO LONGER MINIMUM DIMENSIONS REQUIREMENT

Notes:

- [4.Q] *Are there any specifications on the LED array to make it visible in bright daylight?*
[4.A] No, teams should verify this for themselves. Our tests used an 8x8 NeoPixel matrix but you may use any array you chose to purchase or build. An 8 cm diameter, or 8 x 8 cm square are recommended minimum dimensions. We suggest use of a sunshade to block direct sunlight.

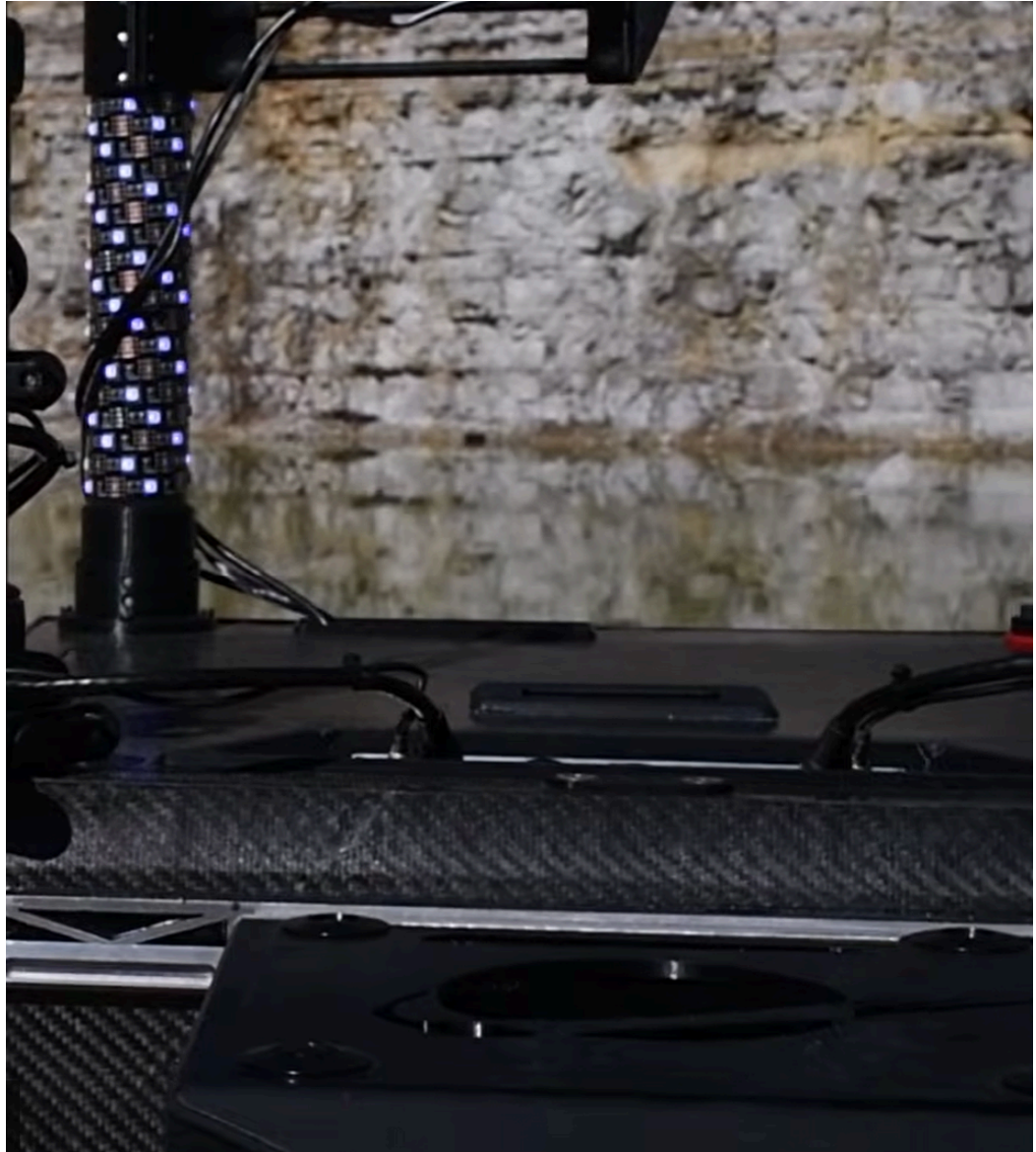
Source: [Requirements & Guidelines](#)

Notes:

- [3.Q] *Are there any specifications on the LED array to make it visible in bright daylight?*
- [3.A] No, teams should verify this for themselves. We suggest use of a sunshade to block direct sunlight, but you may use any array, high power LED or other light you chose to purchase or build. Just make sure it is clearly visible for anyone following the rover to know what state the rover is in.
- What URC requires (from 2026):
 - LED indicator on the back of the rover and visible in bright daylight
 - Red: autonomous
 - Blue: teleoperation/manual driving
 - Flashing green: successful arrival
- What they do not specify:
 - A particular LED brightness or technology
- What they recommend:
 - A sunshade to block direct sunlight

Question: Figure out how to make the lights bright enough during the day

1. Given lumens, #of LEDs * lumens = intensity
2. How Other Teams Did It:
 - a. Missouri S&T Rover



b.

Tldr; I am more confident about this option, but I will continue working on option 2 so we have the option but I think this takes less of our resources in time and money if option 2 doesn't actually function. We can decide on either the 8x8, 16x16, 8x32 as they all have the same luminance value. You can save \$10 getting the 8 cm x 8 cm but the 2022 FAQs warn that the minimum size for an array should be 8 cm x 8 cm.

Note: the nits for blue are lowest. Investing in a sun shade or having the mechanical team make one would greatly increase confidence for when the autonomous mode is running that the blue LEDs will actually be seen.

The panel comes in the following sized options:

- 1) 8 cm x 8 cm
- 2) 8 cm x 32 cm
- 3) 16 cm x 16 cm

Given in datasheet in 'option 1' tab:

Datasheet name	Value
Red brightness, central	360 mcd = 0.360 cd
Green brightness, central	1150 mcd = 1.150 cd
Blue brightness, central	220 mcd = .220 cd
White brightness, central	1710 mcd
RGB channel constant current	15 mA/channel
Red wavelength	620-625 nm
Green wavelength	515-525 nm
Blue wavelength	465-475 nm

8 cm x 8 cm Panel Calculations:

- 1) $N = 8 * 8 = 64$ LEDs
- 2) $\text{Area} = (0.08) * (0.08) = 0.0064 \text{ m}^2$

3) luminance:

a) $I_{\text{red}} = 64 * (0.360) = 23.04$

b) $L_{\text{red}} = I / A = 23.04 / 0.0064 = 3600 \text{ cd/m}^2 =$

c) $L_{\text{red}} = 3600 \text{ nits}$

d) $I_{\text{green}} = 64 * 1.150 = 73.60 \text{ cd}$

e) $L_{\text{green}} = 73.60 / 0.064$

f) $L_{\text{green}} = 11500 \text{ nits}$

g) $I_{\text{blue}} = 64 * 0.220 = 14.08 \text{ cd}$

h) $L_{\text{blue}} = 14.08 / 0.0064 = 2200 \text{ cd/m}^2$

i) $L_{\text{blue}} = 2200 \text{ nits}$

j) $I_{\text{white}} = 64 * 1.710 = 109.44 \text{ cd}$

k) $L_{\text{white}} = 109.44 / 0.0064 = 17100 \text{ cd/m}^2$

l) $L_{\text{white}} = 17100 \text{ nits}$

16 cm x 16 cm Panel Calculations:

1) $N = 16 \times 16 = 256$

2) $A = (0.16)(0.16) = 0.0256 \text{ m}^2$

3) luminance

a) $I_{\text{red}} = 256 * (0.360) = 92.16 \text{ cd}$

b) $L_{\text{red}} = I / A = 92.16 / 0.0256 = 3600 \text{ cd/m}^2$

c) $L_{\text{red}} = 3600 \text{ nits}$

d) $I_{\text{green}} = 256 * 1.150 = 294.40 \text{ cd}$

e) $L_{\text{green}} = 294.4 / 0.0256$

f) $L_{\text{green}} = 11500 \text{ nits}$

g) $I_{\text{blue}} = 256 * 0.220 = 56.32 \text{ cd}$

h) $L_{\text{blue}} = 56.32 / 0.0256 = 2200 \text{ cd/m}^2$

i) $L_{\text{blue}} = 2200 \text{ nits}$

j) $I_{\text{white}} = 256 * 1.710 = 437.76 \text{ cd}$

k) $L_{\text{white}} = 437.76 / 0.0256 = 17100 \text{ cd/m}^2$

l) $L_{\text{white}} = 17100 \text{ nits}$

16 cm x 16 cm Panel Calculations:

4) $N = 8 \times 32 = 256$

5) $A = (0.08)(0.32) = 0.0256 \text{ m}^2$

6) luminance

a) $I_{\text{red}} = 256 * (0.360) = 92.16 \text{ cd}$

b) $L_{\text{red}} = I / A = 92.16 / 0.0256 = 3600 \text{ cd/m}^2$

c) $L_{\text{red}} = 3600 \text{ nits}$

d) $I_{\text{green}} = 256 * 1.150 = 294.40 \text{ cd}$

e) $L_{\text{green}} = 294.4 / 0.0256$

f) $L_{\text{green}} = 11500 \text{ nits}$

g) $I_{\text{blue}} = 256 * 0.220 = 56.32 \text{ cd}$

h) $L_{\text{blue}} = 56.32 / 0.0256 = 2200 \text{ cd/m}^2$

i) $L_{\text{blue}} = 2200 \text{ nits}$

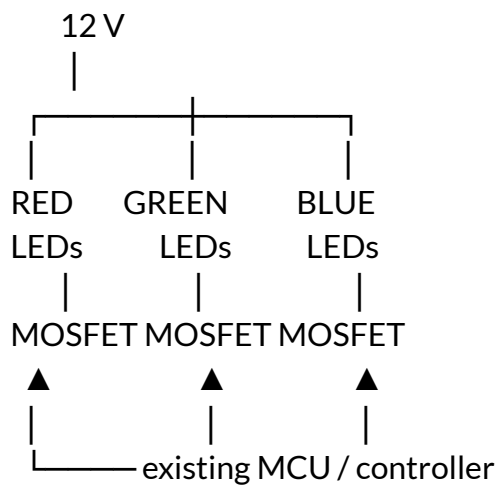
j) $I_{\text{white}} = 256 * 1.710 = 437.76 \text{ cd}$

k) $L_{\text{white}} = 437.76 / 0.0256 = 17100 \text{ cd/m}^2$

l) $L_{\text{white}} = 17100 \text{ nits}$

We could just not do an addressable matrix and build one. The idea: Option 2 is a non-addressable RGB status light. Instead of commanding 64 individual LEDs with a WS2815 data signal, you use a normal 12 V RGB LED module with three channels: red, green, and blue. Each color channel is switched by a MOSFET, and the rover controller only needs to output three simple GPIO signals. This is electrically simpler and directly matches URC's required states: red for autonomous, blue for teleoperation, and flashing green for successful arrival.

Architecture:



Per LED:

$$I_{LED} = 0.3 / 12 = 0.025 \text{ A} = 25\text{mA}$$

8 cm x 8 cm Panel Calculations:

- 1) $N = 64$
- 2) $\text{Max Power} = 64 * 0.3 = 19.2 \text{ W}$
- 3) $I = P/V = 19.2/12 = 1.6\text{A}$

16 x 16 / 8 x 32 Panel Calculations:

- 1) $P = 256 * 0.3 = 76.8$
- 2) $I = 76.8/12 = 6.4 \text{ A}$

Note: the guide suggests 20% - 30% of headroom for power-supply rather than operating at 100%.

For 8x8: minimum 12 V, 1.6 A -> 12 V, 2-2.5 A

For 16x16 / 8x32 12 V, 6.4A -> 12 V, 8 A

Connections —

For the Status LED, my expected connection would be **Battery** → **PDB** → [**12 V buck-boost regulator if needed**] → **WS2815 LED array** for the LED power.

- 1) **Raspberry Pi** would control the array, powered from an appropriate **5 V step-down supply** rather than directly from 12 V.
- 2) Since the Raspberry Pi GPIO uses 3.3 V logic, I would also include a **3.3 V-to-5 V logic level shifter** between the Raspberry Pi data output and the WS2815 data input. The Raspberry Pi, logic shifter, LED array, and PDB would all share a **common ground**. If the PDB already provides a regulated 12.0 V output, then the dedicated 12 V regulator for the LED array may not be necessary.

Hi, I'm reviewing the Status LED power design and wanted to confirm a few things before I look at the regulator. I'm basing this on the current battery CAD and draw.io, but I'm not sure whether that battery design is final. If it is not final yet, that is fine, I'm assuming a nominal 12 V system based on what we talk ab monday.

From the CAD, it looks like there are 24 LiFePO₄ 32700 cells rated 3.2 V, 6 Ah each, maybe configured as 4S6P. The draw.io shows the battery feeding the Power Distribution Board (PDB), with several 12 V branches.

So if that is the final design,

- is it configured as 4S6P?
- What are the minimum and maximum battery voltages during normal operation?
- Are the power distribution board outputs labeled 12 V actually regulated to 12.0 V, or is that a nominal 12 V?
- Does the battery management system regulate the output voltage
- Is there any overall power-budget limitation I should keep in mind when sizing the Status LED/Raspberry Pi branch particularly because the currents are looking different depending on the size of the array (for example 8cm x 8cm needs 1.6A at its max but the 16x16 or 8x32 cm ones I am looking like need at max 6.4 A)

My expected Status LED setup is:

Battery → PDB → [12 V regulator if needed] → WS2815 LED array

and separately:

PDB → 5 V regulator → Raspberry Pi → 3.3 V-to-5 V logic level shifter → WS2815 data input

with the Raspberry Pi, logic level shifter, LED array, and PDB all sharing a common ground.

If the PDB already provides regulated 12 V and 5 V outputs, then I may not need to add separate regulators.